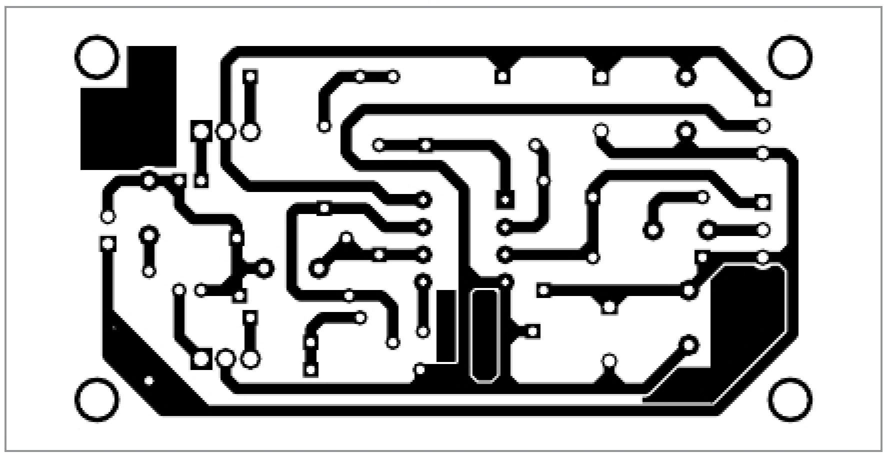


Fig. 2: Actual-size PCB layout of the universal circuit

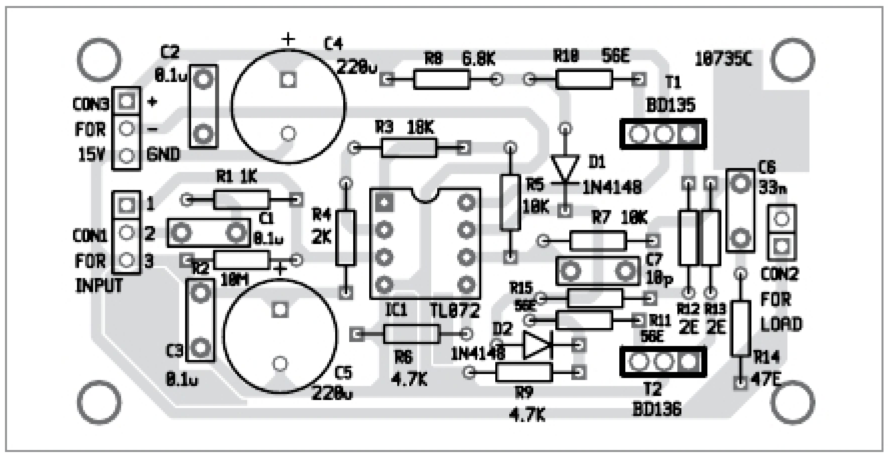


Fig. 3: Components layout for the PCB

tor R15 should have the minimum applicable value to have minimal THD and IMD.

Resistors R12 and R13 operate as current-limiting elements. Maximum output current is determined by the following relationship:

Imax = 0.68V/R12 = 0.68V/R13 = 340mA

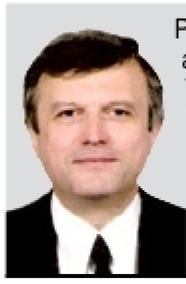
R10, R11, C6 and R14 reduce THD and increase stability of the circuit.

Construction and testing

An actual-size PCB layout of the circuit is shown in Fig. 3 and its components layout in Fig. 4. After assembling the circuit on the PCB, connect a ±15V power supply across CON3.

You can give three types of inputs—DC, AC and audio—across CON1. Connect DC input across terminals 1 and 3 of CON1, or AC input across terminals 2 and 3 of CON1. Connect audio signal input from the mobile/laptop/PC across terminals 2 and 3 of CON1.

The circuit can be tested with a resistive load of 100-ohm connected across CON2 using a power supply of ±15V and AC input signal of ±1Vpp, which should give a total gain of -10. The circuit does not require any adjustment. You may need small heat-sinks for T1 and T2 with thermal resistance around 35°C/W. EFY



Petre Tzv Petrov was a researcher and assistant professor in Technical University of Sofia, Bulgaria and expert-lecturer in OFPPT (Casablanca), Kingdom of Morocco. Now he is working as an electronics engineer in the private sector in Bulgaria